

Cooper Lopis

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EDUCATION

B.S. Electrical Engineering University of Colorado Boulder, CO '28

EXPERIENCE

Process Controls Engineer Intern PIM Brands – Somerset, NJ *Jun 2025 – Aug 2025*

Global confectionery manufacturer producing Welch's® Fruit Snacks, Sour Jacks®, and other candy brands.

- Led an intern project initiative to design a new autonomous menu system cutting changeover times by an estimated 30 minutes
- Designed and deployed SCADA interfaces in Ignition to monitor and control industrial process control systems
- Wrote SQL queries to extract and analyze production data
- Gained hands-on experience with the full engineering process: requirements gathering, design, testing, and documentation

Engineering & Facilities Intern Town Square Innovative Adult Daycare – Fair Lawn, NJ *May 2025 – Sep 2025*

- Assisted in the engineering and buildout of a 10,000 sq. ft. facility, collaborating with licensed engineers and electricians on wiring, alarm systems, and automated door installations per municipal and state code
- Contributed to the design and installation of all integrated hardware throughout the facility
- Provided feedback and support to the development team during layout and installation phases

Executive Assistant to President/CEO Visiting Angels of Fort Lee – Fort Lee, NJ. *May 2021 – Present*

In home healthcare provider serving seniors and adults who need assistance with daily living.

- Led digital transformation initiative: migrated 15 years of paper records to a fully automated electronic platform
- Organized and maintained client and employee files, ensuring data accuracy across the company's updated software system
- Supported executive operations and business development over 5+ years, contributing to measurable agency growth

Certified Lifeguard Indian Trail Club – Franklin Lakes, NJ. *May 2022 – Aug 2022*

- Held CPR certification and Red Cross lifeguard certification; monitored pools, lake, group swims, and swim meets

PROJECTS

Arduino-to-FPGA Digital Voltmeter 2024

- Designed a digital voltmeter bridging an Arduino microcontroller and an FPGA development board, programming in Verilog to drive a 7-segment hex display
- Core challenge: precisely matching clock rates and implementing reliable UART data transmission between the two platforms to achieve accurate voltage readings

Ham Radio Build & License 2024

- Built a functional radio, receiver chain and transmitter chain, from scratch and earned an FCC Amateur Radio (Technician) license
- Deepened understanding of RF signal propagation, antenna tuning, and analog circuit design through hands-on construction

2D Platformer Game (Unity / C#) Ongoing

- Developing a fully-featured 2D platformer in Unity with C#, implementing advanced movement mechanics and various interconnected systems

TECHNICAL SKILLS

Languages: C#, Python, C, Verilog, SQL

Tools & Platforms: Unity, Ignition SCADA, Git, Arduino

Concepts: SCADA Design, PLC Logic, UART Communication, FPGA Programming, OOP, Digital Logic

Certifications: FCC Amateur Radio License (Technician), ARC Lifeguard & CPR Certified